

Design Rule Extraction Using Multi-Fidelity Surrogate Model for Unmanned Combat Aerial Vehicles

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Due to the instability of the pitching moment induced by the strong vortical flow at moderate to high angles of attack in the unmanned combat aerial vehicle (UCAV) configuration, previous design studies focused primarily on low-speed flight conditions where the longitudinal stability is a serious issue. Furthermore, because of the limitation of computational cost, univariate analyses based on low-fidelity flow solvers are performed to investigate the relationships between design variables and quantities of interest. In this study, since the high-speed condition may occupy a high proportion in the UCAV mission, it was additionally considered through multi-objective optimization. A high-fidelity flow solver is used for both low-speed and high-speed conditions to accurately capture the vortical flow structure, and corresponding computational costs are relieved by multi-fidelity surrogate modeling. Based on this surrogate model, design rules are extracted, and the effects of design variables are investigated through univariate and multivariate data mining techniques. At last, universal design rules that can be referenced regardless of the parameterization method are visualized in an airfoil shape using proper orthogonal decomposition. The time-efficient design rule extraction framework and the approach to analyzing extracted design rules presented in this study can be extended to any engineering field.

Nomenclature

C_D	= drag coefficient
C_L	= lift coefficient
C_M	= pitching-moment coefficient
C_P	= pressure coefficient
D	= drag, N
d_{ci}^2	= distance between the winner and the neighboring node in self-organizing map
E_j	= energy ratio of top j modes in proper orthogonal decomposition
F	= kriging regression matrix
$h_{ci}(i)$	= Gaussian function at i th iteration in self-organizing map
L	= lift, N
R	= kriging correlation matrix
$R_{L.E.}^1$	= leading-edge radius at section1 (nondimensionalized by airfoil chord length)
$R_{L.E.}^2$	= leading-edge radius at section2 (nondimensionalized by airfoil chord length)
r	= kriging correlation vector
X_{low}^1	= x coordinate of the lower crest at section1 (nondimensionalized by airfoil chord length)
X_{low}^2	= x coordinate of the lower crest at section2 (nondimensionalized by airfoil chord length)
X_{up}^1	= x coordinate of the upper crest at section1 (nondimensionalized by airfoil chord length)
X_{up}^2	= x coordinate of the upper crest at section2 (nondimensionalized by airfoil chord length)
x_{hf}	= input of high-fidelity sample data in hierarchical kriging
x_{lf}	= input of low-fidelity sample data in hierarchical kriging
y_{hf}	= output of high-fidelity sample data in hierarchical kriging

$\hat{y}_{hf}$	= high-fidelity kriging model in hierarchical kriging
$\hat{y}_{lf}$	= low-fidelity kriging model in hierarchical kriging
Z_{lf}	= Gaussian random processes of low-fidelity kriging model in hierarchical kriging
Z_{low}^1	= z coordinate of the lower crest at section1 (nondimensionalized by airfoil chord length)
Z_{low}^2	= z coordinate of the lower crest at section2 (nondimensionalized by airfoil chord length)
Z_{up}^1	= z coordinate of the upper crest at section1 (nondimensionalized by airfoil chord length)
Z_{up}^2	= z coordinate of the upper crest at section2 (nondimensionalized by airfoil chord length)
$\beta_{0,hf}$	= coefficient of trend model for high fidelity in hierarchical kriging
$\beta_{0,lf}$	= coefficient of trend model for low fidelity in hierarchical kriging
θ_{twist}^1	= twist angle at section1, deg
θ_{twist}^2	= twist angle at section2, deg
λ_i	= eigenvalue of the i th mode in proper orthogonal decomposition
$\hat{\mu}_i(x_i)$	= variance component of objective function due to i th design variable in analysis of variance
$\hat{\mu}_{total}$	= total variance of objective function in analysis of variance
ρ	= Spearman's rank correlation coefficient
$\sigma(i)$	= radius of the neighborhood at i th iteration in self-organizing map

I. Introduction

UNMANNED combat aerial vehicles (UCAVs) are capable of performing suppression of enemy air defenses and intelligence, surveillance, and reconnaissance missions, and offer several advantages, such as high maneuverability and G-force endurance, the ability to operate in high-risk environments, and fewer design constraints owing to the absence of a cockpit [1–3]. Considering the mission profile of UCAVs, they should be capable of efficient long-range and long-endurance flight to penetrate enemy lines with aircraft ordnance. In addition, they also need low observability to remain undetected by enemy radar during infiltration. Accordingly, a blended wing body design is generally adopted for UCAVs as it reduces the total drag, radar cross section, and noise levels [4,5]. However, the tailless configuration poses significant challenges in maintaining aerodynamic stability [6]. As the control surfaces are located at the trailing edge of the wing, it is difficult to maintain

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longitudinal stability due to a short moment arm and the fact that a larger force is required to generate a given pitch change [7]. Consequently, UCAVs generally use backward swept wings to alleviate this problem [8].

UCAVs with swept wings exhibit nonlinear aerodynamic characteristics due to the formation of complex leading-edge vortices at moderate to high angles of attack (AOA) and the corresponding flow separation [9,10]. As the AOA increases, flow separation occurs near the wingtip due to the vortex generated at the leading edge, thereby reducing the suction effect behind the moment reference point (MRP) and inducing a nose-up moment. Accordingly, the pitching moment increases nonlinearly with the increase in AOA, which is called pitch break [11]. Pitch break is a chronic problem for UCAVs, especially at low speed, at which a larger lift is required [12–14]. As this phenomenon makes the control surfaces be in the separated flow region, the controllability of the aircraft deteriorates [12]. Consequently, the maximum usable lift coefficient is reduced, requiring a higher rotational speed at takeoff and a larger and heavier engine [15,16]. Therefore, the low-speed longitudinal stability should be considered during the early design stages to save development costs.

Numerous studies have focused on the accuracy of flow analysis to investigate the aerodynamic characteristics of UCAVs. McParlin et al. [12] analyzed the experimental results of the 1303 UCAV concept at a Mach number range of 0.13–0.3, considering various leading-edge radii. Computational analyses have also been conducted by several researchers. Schütte [17] investigated the aerodynamic characteristics of swept wings with various leading-edge radii and sweep angles. Liersch et al. [18] performed a parametric study of the MULDISCON (multidisciplinary configuration) UCAV, considering various airfoil shapes and leading-edge radii. They assumed a Mach number of 0.4 for the flow simulation, which represents a computational compromise between low-speed and high-speed flight. For the same configuration and flight conditions, Schütte et al. [19] considered additional variations in the twist angle and MRP location. These simulations were performed using the Reynolds-averaged Navier–Stokes (RANS) solver to evaluate the pitching-moment characteristics. However, due to the enormous computational costs, these analyses were limited to narrow design space and univariate parametric studies, wherein only one variable varies while the others remain fixed.

Other studies have explored a large design space through optimization framework in the early design phase. Nguyen et al. [20] and Tyan et al. [21] performed variable-fidelity optimization for various UCAV planforms. They used empirical relations and an extended vortex lattice solver to calculate the static margin (SM) for stability & control (S&C) analysis. Karakoc and Kaya [22] set the leading-edge radius and twist angle as design variables in the MULDISCON UCAV and performed optimization increasing the maximum usable lift coefficient at low-speed conditions. However, though they explored vast design spaces through optimization frameworks, the multivariate data mining to investigate the physics between design variables and quantities of interest (QOIs) was not performed. Furthermore, they were either limited to low-fidelity solvers (Refs. [20,21]) or low-speed flight conditions (Ref. [22]), indicating the significant computational costs of high-fidelity flow solvers considering both low-speed and high-speed flight conditions.

These studies implied the practical challenges associated with exploring a large design space for both low-speed and high-speed flight conditions using a high-fidelity flow solver. Consequently, previous studies on UCAV designs have limitations in that there is no study that addresses the following issues simultaneously:

- 1) Flow analysis was limited to low-speed conditions, and high-speed performance was not considered.
- 2) Multivariate analysis using high-fidelity aerodynamic solver was not performed in scrutinizing the physics between the design variables and QOIs.

This study attempts to overcome these limitations comprehensively. Although the longitudinal stability is crucial at low-speed flight conditions, the high-speed performance should also be considered during the design process because it may account for a high proportion in the UCAV mission as in Ref. [10]. Furthermore, Nangia et al. [23] argued that the aerodynamic design of a UCAV should

compromise between low-speed stability and high-speed performance. Accordingly, both flight conditions are taken into account herein. Also, an adequate fidelity solver should be selected for accurate and efficient S&C analysis. Tomac et al. [24] compared three engineering methods for the S&C analysis of UCAVs and figured out that though the analyses were performed in a linear flow-physics region, they failed to predict the pitching-moment coefficients measured in wind-tunnel experiments. Moreover, when compared to the experimental data, Petterson [13] showed that the pitching-moment characteristics obtained using the Euler solver were overall poor and the location of the first pitch break was missed, whereas Hall et al. [25] demonstrated that the pitching moment could be successfully calculated using RANS analysis with a moderate grid resolution when compared to the experimental data. Therefore, this study uses a RANS solver as a high-fidelity tool to analyze both flight conditions simultaneously: more specifically, a multi-objective optimization framework is adopted to consider both aerodynamic QOIs (the first objective is high-speed cruise performance, and the second is low-speed longitudinal stability).

However, high-fidelity flow analyses at these two flight conditions with various AOA require enormous computational cost. Herein, its cost is relieved by multi-fidelity modeling (also called variable-fidelity modeling, VFM) [26]. It is surrogate-based modeling that fuses data from different information sources (usually having disparate fidelities) to efficiently construct a surrogate model and has been widely used in aerospace engineering due to its potential to reduce high-fidelity computation (for example, Kaya et al. [27] showed that for the MULDISCON UCAV configuration, the aerodynamic coefficients with respect to AOA can be predicted efficiently and accurately through cokriging-based VFM). In this regard, the hierarchical kriging (HK) surrogate model [28] with an adaptive sampling method is employed in this study to ensure computational efficiency of the flow analysis.

Furthermore, this study also focuses on the limitations of the previous studies that performed only univariate parametric analyses on the aerodynamic characteristics (e.g., Schütte et al. [19] investigated the behavior of various aerodynamic coefficients at different leading-edge radii, while keeping other variables such as the twist angle and MRP fixed). However, univariate analyses are inherently limited as they are not able to analyze how QOI changes with multiple design variables. While the univariate analysis can be a good option for a given computational resource, design rules extracted from it should not be unconditionally referenced in the design process. In the real world, design variables affect aerodynamics in a multivariate manner: they have intricate and entangled relationships, therefore the results of the multivariate analysis can be totally different from the univariate analysis. In addition, design-space exploration based on univariate analyses do not guarantee the exploration of the region where the optimal solutions actually exist. In this context, design studies based only on univariate analyses cannot offer practical and reliable design guidance for future preliminary design studies. Herein, both univariate and multivariate analyses are performed to investigate the complex but realistic design rules between the design parameters and the aerodynamic QOI from both flight conditions. Sample designs are generated from the constructed VFM surrogate model, and various data mining techniques are applied to them. The results of these data mining techniques are scrutinized in three categories: univariate analyses, multivariate analyses, and airfoil shape visualization by proper orthogonal decomposition (POD). Both univariate and multivariate analyses are conducted for efficient and practical design rule extraction. To be more specific, sensitivity analysis, analysis of variance (ANOVA), and self-organizing maps (SOMs) are used for the univariate analyses. For the multivariate analyses, decision tree and rough set theory are applied. By comparing the results of the methods belonging to these two categories (univariate and multivariate), it is confirmed that the design rules extracted from them can be different, which emphasizes the motivation of this study. Finally, the universal design rules that can be referenced regardless of the parameterization methods are visualized in an airfoil shape using POD. This time-efficient design rule extraction framework based on VFM and the approaches to examining

extracted design rules presented in this study can be extended to any engineering discipline.

II. Methodologies

A. Flow Analysis

KFLOW, a parallelized multiblock-structured RANS tool, is used for flow analysis herein [29–31]. To validate the KFLOW computational fluid dynamics (CFD) solver, the stability and control configuration (SACCON) UCAV was used. The flow conditions used to compare the pressure distribution calculated using KFLOW and the experimental data [14] are Mach number of 0.15 and Reynolds number of 1.6×10^6 . Considering a situation wherein the influence of the vortex is dominant, a high AOA (15.3 deg) was selected. Roe's flux-difference splitting [32] and a second-order Van Leer limiter [33] were used for the inviscid term. The central differencing method and the diagonalized alternating direction implicit method [34] were applied for the viscous term and steady-state solutions, respectively. For simulations of turbulent flows, the $k-\omega$ shear stress transport (SST) model [35] was employed as Hall et al. [25] demonstrated that the predicted pitching moment of the 1303 UCAV concept using this model was in good agreement with the experiment results. A total of 12.5 million volume cells comprising $159 \times 221 \times 360$ grid points and satisfying an average y^+ value of less than 1 were generated for the validation process. As yaw and roll angles are not considered, symmetric flow was assumed for computational efficiency. Since Tomac and Stenfelt [8] successfully predicted the experimental aerodynamic characteristics using a steady RANS solution, a steady-state simulation was performed herein. Note that the CFD analyses during the subsequent optimization processes are performed using the same settings.

The cross sections of the SACCON UCAV (without sting) that was used to compare the calculated pressure distribution and the experimental values are based on Ref. [14], so the locations and designations of these sections can be checked here. The C_p calculated using the KFLOW solver is compared with the experimental values in Fig. 1. The calculation results using the Euler solver are also shown to confirm that the inviscid solution cannot predict the vortical flow accurately. The pressure distribution calculated using the RANS solver with the $k-\omega$ SST turbulence model is similar to the experimental results, whereas the vortical flow predicted by the Euler solver differs significantly (it can be judged that the Euler solver

predicted the wrong vortex topology since it computed the wrong suction peak [14]). Thus, considering the poor performance of the Euler solver, the RANS solver should be used for stability analysis despite its higher computational costs (note that the additional flow solver validation comparing C_L , C_D , and C_M with respect to AOA was also conducted: their results are shown in the Appendix).

B. Variable-Fidelity Modeling

Surrogate-based optimization (SBO) is commonly used in aerodynamic optimization studies for efficient global optimization (EGO) [36,37]. Unlike direct optimization, SBO uses surrogate models to reduce the evaluation of expensive computing simulations and provides high efficiency in the high-dimensional design space where the curse of dimensionality occurs. There are several types of surrogate models that can be used for SBO; the kriging model [38] is commonly used as it can describe both multivariate and nonlinear functions [39,40]. In particular, it provides variance of the predictor, which can be used for adaptive sampling processes such as the expected improvement (EI) in EGO [41,42]. Instead of computing all the cases based on single-level, high-fidelity CFD simulations to generate a surrogate model, variable-fidelity modeling (VFM) uses several fidelity levels to reduce computational expenses.

A hierarchical kriging (HK) surrogate model was employed for the variable-fidelity SBO in this study. It combines the advantages of existing VFM methods: the simplicity and robustness of the correction-based VFM and the accuracy of the cokriging-based VFM [28]. It first constructs a low-fidelity kriging model [Eq. (1)] that is used as a trend function in the high-fidelity kriging model, as shown in Eq. (2):

$$\hat{y}_{lf}(\mathbf{x}_{lf}) = \beta_{0,lf} + Z_{lf}(\mathbf{x}_{lf}) \quad (1)$$

$$\hat{y}_{hf}(\mathbf{x}_{hf}) = \beta_{0,hf} \hat{y}_{lf}(\mathbf{x}_{hf}) + \mathbf{r}^T(\mathbf{x}_{hf}) \mathbf{R}^{-1}(\mathbf{y}_{hf} - \beta_{0,hf} \mathbf{F}) \quad (2)$$

As the kriging model of the higher fidelity level is constructed with an assist from the lower-fidelity kriging model, this model is called the hierarchical kriging model. Detailed mathematical formulas of the HK model can be found in Ref. [28].

This HK model has been employed in several studies owing to its time efficiency and accuracy. Kim et al. [43] observed that for the film-cooling hole array optimization to minimize the average temperature of the nozzle pressure side, the CPU time required for HK-based optimization using two fidelity levels of grid systems was 40% less than the optimization with single-fidelity kriging. Wilke [44] indicated that an HK model with three fidelity levels was 83% more efficient than a single-fidelity model for rotor-blade optimization. Han et al. [45] demonstrated that the two-level HK model provided a time savings of 63 and 75% for the optimization of the ONERA M6 wing and NACA 0012 airfoil, respectively. He also discovered that the estimated optimal solutions were more precise and superior compared to a single-level fidelity kriging model. The details for implementing this efficient and accurate model will be described in Sec. III.

C. Data Mining

1. Univariate Analyses

Two data mining techniques used to scrutinize the univariate relationships between the design variables and the objective functions were demonstrated in this section: ANOVA and SOM. First, an ANOVA can estimate the effect of each design variable on the QOI by decomposing the total variance of the QOI. Specifically, the total variance of the objective function is decomposed into the effects of each design variable x_i as follows:

$$\hat{\mu}_i(x_i) \equiv \int_n \cdots \int_1 f(x_1, \dots, x_n) dx_1 \cdots dx_{i-1} dx_{i+1} \cdots dx_n - \hat{\mu}_{\text{total}} \quad (3)$$

where

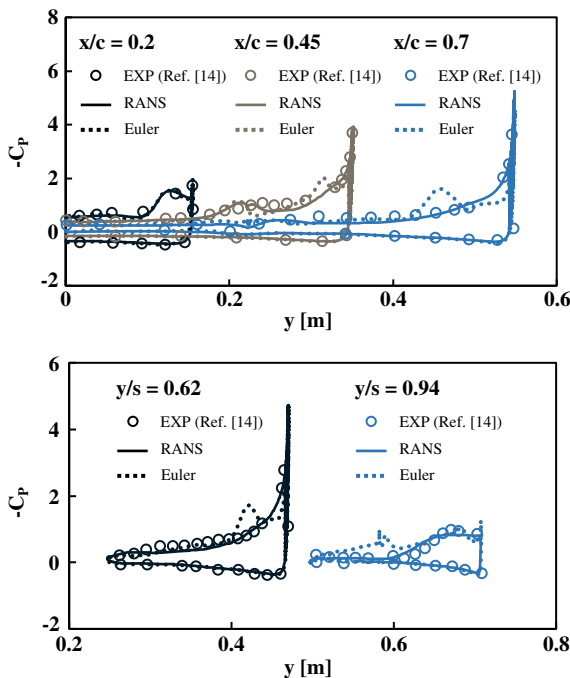


Fig. 1 Comparison of C_p distribution at $Ma = 0.15$, $Re = 1.6 \times 10^6$, $AOA = 15.3$ deg from KFLOW and experimental results.

$$\hat{\mu}_{\text{total}} = \int_n \cdots \int_1 f(x_1, \dots, x_n) dx_1 \cdots dx_n \quad (4)$$

Finally, the influence of x_i on the output is quantified as follows:

$$P = \hat{\sigma}_{x_i}^2 / \hat{\sigma}_{x_{\text{total}}}^2 \quad (5)$$

where

$$\begin{aligned} \hat{\sigma}_{x_i}^2 &= \int_i \{\hat{\mu}_i(x_i)\}^2 dx_i \quad \text{and} \\ \hat{\sigma}_{\text{total}}^2 &= \int_n \cdots \int_1 \{f(x_1, \dots, x_n) - \hat{\mu}_{\text{total}}\}^2 dx_1 \cdots dx_n \end{aligned} \quad (6)$$

More detailed mathematical formulas for this decomposition can be found in Ref. [46]. As ANOVA enables a quantitative examination of the effect of design variables on the objective QOI by the integration within the continuous design space [47,48], it is commonly used for the overall design space exploration.

The SOM is a dimension-reduction technique that projects high-dimensional space into two-dimensional (2-D) space without any information loss [49,50]. This makes it easy to visualize the relationships between variables, in this case, between the design variables and the objective functions. SOM is based on an unsupervised artificial neural network and is trained to project similar input data onto the neighboring area of a 2-D output map. Its learning process consists of three parts: competition, cooperation, and adaptation. In the competition process, each output node in the output map has its own weight vector and competes with other nodes to represent the original data in terms of the weight vector as accurately as possible. The node closest to the original data is determined to be the winner. In the cooperation process, learning is performed using the neighborhood function such that the nodes near the winner have similar characteristics to each other. Consequently, neighboring nodes have an organic relationship with each other. These characteristics provide the basis for qualitatively judging the relationship of each variable based on the contour of the output nodes. In this study, the Gaussian function is selected as the neighborhood function and its expression at i th iteration is as follows:

$$h_{ci}(i) = \exp(-d_{ci}^2 / 2\sigma^2(i)) \quad (7)$$

where d_{ci}^2 is the distance between winner and neighboring node, and $\sigma(i)$ denotes the radius of the neighborhood that determines the effective range of the winner node. Finally, in the adaptation process, the weight vectors of the winner and its nearby nodes are updated iteratively to obtain similar information to the original high-dimensional data. For more detailed mathematical formulas of SOM, see Ref. [49].

2. Multivariate Analyses

Two data mining techniques are applied for the multivariate analysis of sample designs: decision tree and rough set theory. The decision tree machine-learning technique divides the existing dataset to maximize the difference in variance between the separated groups [51]. Herein, a regression decision tree is used to group the used data points according to each QOI. Based on these divided groups, design rules are extracted in the direction of the increasing value of the objective function as our purpose is to maximize its value. This simple algorithm allows easy-to-understand design rules to be extracted based on the divided groups, enabling efficient design rule mining [52,53].

Rough set theory was developed to generate IF-THEN rules from imprecise and ambiguous data [54]. Unlike decision trees that extract a single design rule, rough set theory is particularly powerful for multivariate data mining in that it can extract multiple design rules [52,55]. Before the analysis, the design variables and objective functions of the sample points are categorized. They are divided into three levels depending on their values, such that an equal number of samples exists in each group: level1 (low), level2 (medium), and level3 (high). Herein, each sample used in the rough set analysis

consists of a set of design variables and objective functions, and it is called a rule. However, there can be rules that are indistinguishable with respect to some design variables since those variables have a negligible effect on the objective function. To eliminate these redundant rules, a reduction process follows to select characteristic decision-making features. In this process, it extracts the conditions of the critical design variables (IF-part) that make the value of the objective function high as level3 (THEN-part). By this reduction technique, precise and straightforward design rules can be extracted by discarding design variables that are unnecessary or have a negligible influence on increasing the objective function. Finally, the reduced IF-THEN rules are filtered to ensure the ease and efficiency of the analysis. Detailed mathematical formulas for deriving the rough sets are provided in Ref. [54].

3. Proper Orthogonal Decomposition

POD is a dimensional reduction technique that projects the original data onto a set of orthogonal vectors [56] and it is applied herein to extract design rules from Pareto solutions as shown in Ref. [57]. First, orthogonal vectors used for the projection are selected to maximize the variance of the projected data as follows:

$$\mathbf{e} = \arg \max_{\mathbf{e}} (\text{Var}(\mathbf{Me})) \text{ s.t. } \|\mathbf{e}\| = 1 \quad (8)$$

The constrained optimization equation, Eq. (8), is approximated using Lagrangian relaxation and can be transposed to the eigenvalue-decomposition problem. In this study, singular value decomposition is used for computational efficiency.

The data used as the POD input are selected from the airfoil shape of the designs to be data mined. The airfoil is discretized so that z coordinates of the fixed x coordinates are extracted. Accordingly, $100 \times z$ coordinates (z) are used as the input of each airfoil. Subsequently, each airfoil coordinates vector of the i th Pareto solution $\mathbf{z}(i)$ is decomposed into eigenvectors as follows:

$$\mathbf{z}(i) = a_1(i) * \mathbf{m}_1 + a_2(i) * \mathbf{m}_2 + \cdots + a_{100}(i) * \mathbf{m}_{100} \quad (9)$$

where $\mathbf{m}_j$ is the eigenvector of the j th mode and $a_j(i)$ represents the component of j th mode and i th Pareto solution. The energy ratio of the top j modes is calculated as

$$E = \sum_{k=1}^j \lambda_k / \sum_{k=1}^{100} \lambda_k \quad (10)$$

where λ_k is the eigenvalue of the k th mode. The energy ratio E is used as the basis for determining the number of modes to be used in subsequent analyses. These overall POD processes are similar to those in Ref. [57].

III. Results and Discussion

A. Variable-Fidelity Optimization

Although the HK model can be used with any number of fidelity levels, only two are used herein: low-fidelity (LF) and high-fidelity (HF). The type of fidelity can be selected based on the physics model (e.g., Euler or RANS solver) or the degree of discretization (e.g., coarse or fine grid) [58]. In this study, the grid density is adopted as the fidelity criterion. A grid convergence test was performed to determine the appropriate density of the LF and HF grids. A full-scale model of the X-47B configuration, a double delta-wing UCAV designed by Northrop Grumman (Fig. 2), was used as the baseline for this test and subsequent optimization. A double delta-wing configuration offers several advantages over a generic delta-wing configuration (e.g., 1303 UCAV concept): 1) increased length for propulsion and weapons integration; 2) excellent aerodynamic efficiency and carrier suitability; 3) outer wing panels decoupled from the center body (which enables planform optimization); and 4) foldable cranked outer portions for small carrier decks [59]. Clean geometry without engines is used, considering Liersh et al. [18] demonstrated the lift and pitching moments have little dependency on whether or not the

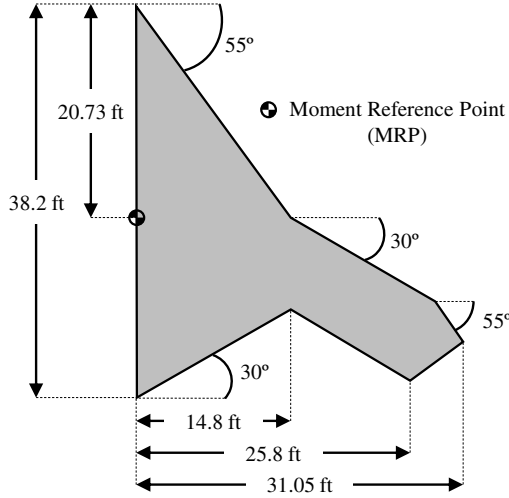


Fig. 2 Top view of full-scale X-47B configuration.

engine is integrated. A total of six structured grids with 2 ~ 12.5 million volume cells with a y^+ value of 1 were generated for this configuration. The flow parameters are as follows: at sea level altitude, AOA as 0 deg, Mach number as 0.15, Reynolds number as 9.12×10^6 , reference length as 2.6725 m, and reference area as 43.714 m². The resultant C_L , C_D , C_M , and CPU time required for various grids are shown in Fig. 3. As the number of grid cells increases, the aerodynamic coefficients exhibit a gradual convergence (the convergence of the used grids in the grid convergence test were quantitatively checked with the grid convergence index as in Ref. [43], and this index indicated they were converged sufficiently). An LF and HF grid density of 2 million and 10 million cells, respectively, were selected for the VFM optimization process (to examine the accuracy of the LF grid, a sensitivity test was conducted in the design space to be determined later, and its trend was judged to be suitable for assisting the HF grid when constructing the HK model). Their corresponding grid topologies are shown in Fig. 4 and they will be used in the later variable-fidelity optimization.

As the fidelity criterion was established, the optimization problem needs to be defined. As PARSEC airfoil parameters are closely related to the aerodynamic characteristics [60,61], the PARSEC parameters of the airfoil at two sections were selected as the design variables, and the twist angles at these sections were considered as well (the purpose of selecting these two sections was to modify the outer wing part effectively. Especially, section2 was chosen to maintain the modified wingtip concept of the X-47B, and more details can be found in Ref. [59]). The corresponding 12 design variables are shown in Fig. 5 (there are five PARSEC parameters and one twist

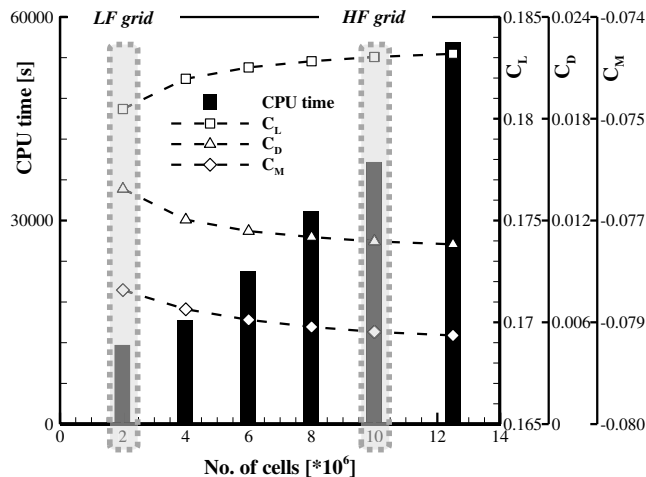


Fig. 3 Grid sensitivity test results comparing C_L , C_D , C_M , and CPU time for 2–12.5 million volume cells.

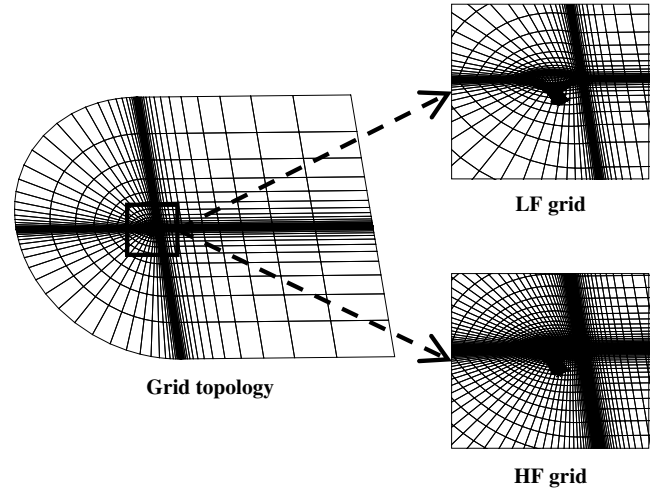


Fig. 4 Overall grid topology (left), LF (2 million volume cells, top right), and HF (10 million volume cells, bottom right) grids around the surface mesh.

angle at both sections so that a set of these 12 design variables determines one UCAV design). The NACA 64A210 airfoil was used as the baseline airfoil, and the design space with respect to the baseline airfoil is as follows for both sections: -50 to $+50\%$ for $R_{L,E}$, -7 to $+7$ deg for θ_{twist} , -30 to $+30\%$ for X_{low} , -10 to $+3\%$ for X_{up} , -3 to $+10\%$ for Z_{low} , and -15 to $+30\%$ for Z_{up} . All design variables are linearly scaled to lie between 0 and 1 and are used in subsequent sections for easy comparison on the same scale.

To consider cruise performance and low-speed longitudinal stability simultaneously, two objective functions were defined. The first objective (OBJ1) is to maximize the value of L/D at cruise and the second objective (OBJ2) is to maximize the static margin (SM) at low speed (to investigate the precise pitch break phenomenon, it is necessary to perform flow analysis from low to high AOA range. However, RANS simulation on various UCAV designs with AOA sweep requires unrealistic computational resources. Accordingly, SM at takeoff flight condition was adopted as a quantitative criterion to represent low-speed longitudinal stability of the UCAV in this study).

OBJ1(L/D_{cruise}): maximize L/D at cruise

OBJ2(SM_{takeoff}): maximize $SM[\%]$ at takeoff

The criterion for the low-speed longitudinal stability, SM (OBJ2), is calculated as

$$SM = -d(C_M)/d(C_L) = -d(C_M)/d(AOA) * d(AOA)/d(C_L) \quad (11)$$

To compute this value, the appropriate AOA for the required C_L at takeoff flight (Table 1) should be found. Therefore, the flow calculation should be repeated in an iterative manner, changing the AOA until the adequate AOA generating specified lift is found (the required lift at each flight condition was estimated based on Raymer's weight estimation method [62]). When the corresponding AOA is found, the SM is estimated using a forward difference scheme (a similar procedure for calculating SM can be found in Ref. [21]). Flight conditions for cruise and takeoff in optimization are summarized in Table 1.

The pitching moment at cruise was set as a constraint (CON) to prevent excessive trim drag while trimming the aircraft:

$$\text{CON}(C_{M,\text{cruise}}): |C_{M,\text{cruise}}| \leq |C_{M_{\text{baseline}}}|$$

On the basis of the selected fidelities and defined optimization problem, initial samples were generated using the design of experiments

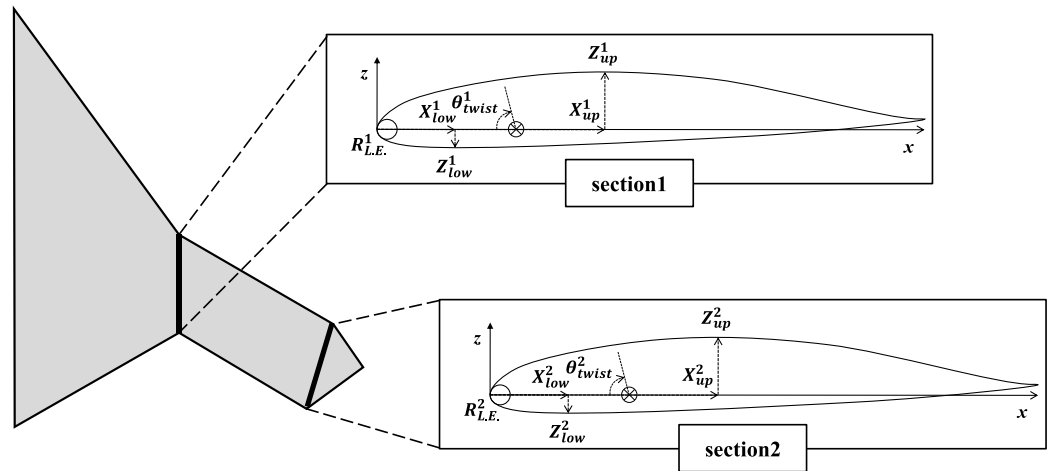


Fig. 5 Design variables for optimization: five PARSEC parameters and twist angle were selected as variables for each section.

Table 1 Flight conditions for optimization: high-speed cruise and low-speed takeoff

Flight conditions	Altitude	Mach number	Reynolds number	Required C_L
Cruise	11 km	0.7	1.4164×10^7	0.1825
Takeoff	At sea level	0.2	1.2447×10^7	0.5248

method. The Latin hypercube sampling (LHS) method was selected owing to its uniformity in the design space [63]. A total of 114 points for the LF model and 20 points for the HF model were independently extracted using LHS (according to the principles of the HK, sampling points for the LF and HF do not need to be related). All points extracted regardless of the fidelity to which they belong were analyzed under both cruise and takeoff flight conditions to obtain values for the two objective functions (L/D_{cruise} and SM_{takeoff}). Subsequently, the initial HK model was constructed based on the QOI of these initial sample points.

Next, a global optimization method should be selected to obtain the Pareto solutions based on the constructed surrogate model. In this study, the genetic algorithm (GA) was selected for its performance in discontinuity and multimodality problems [64]. It is also efficient in multi-objective problems as it performs a parallel search for optimal solutions from the populations. Therefore, the multi-objective genetic algorithm (MOGA) was used herein to calculate the Pareto solutions and the corresponding tradeoff relationship between both objective functions [65]. In particular, the MOGA using Fonseca and Fleming’s [66] Pareto ranking method was adopted. After the crossover and mutation process, the best-N and standard archiving techniques were used to efficiently extract the Pareto solutions, as described in Ref. [65].

The MOGA process coupled with adaptive sampling was repeated until a certain convergence criterion of the surrogate model was satisfied; it terminated when the amount of change in convergence

metric for Pareto solutions (the convergence metric of the current Pareto solutions is calculated with respect to those of the previous iteration: see Ref. [67] for more details) becomes less than 0.5% of the metric value in the previous iteration, which indicates that the additional iterations will have a negligible effect in the subsequent data mining results. For the convergence, adaptive sampling was performed considering the predicted value (PV) and expected improvement (EI) criteria [68]: PV and EI were used together in that they improve the accuracy and uncertainty of the current optimal solutions, respectively (distinct Pareto solutions in terms of PV and EI were calculated with MOGA, and the points to be newly added are selected from these two sets of solutions). This process was particularly crucial in this study as the sample designs to be used for the data mining are extracted in the surrogate model; sample designs from the uncertain surrogate model cause uncertainty in the data mining results based on them. Therefore, adaptive sampling was conducted to ease this uncertainty. It was used only for the HF, and the HK surrogate model converged after adding 13 points to the initial HF sample points. Therefore, the final number of sample points were: 114 points for the LF model and 33 points for the HF model. The corresponding total CPU time using Intel(R) Xeon(R) CPU E5-2690 v3 @ 2.60 GHz (12 cores) was 1468 and 3058 hours for low-fidelity and high-fidelity flow analyses, respectively. The resulting HK surrogate model was cross-validated [41] to confirm its validity. As shown in Fig. 6, most of the points were close to the ideal lines for the two objectives and one constraint. HK, MOGA, and cross-validation were conducted using in-house code, and its algorithms and implementations can be found in Refs. [47,64,65].

Finally, a total of 1281 sample points were selected from the converged HK surrogate model: 300 from the Pareto solutions and 981 from the populations of the last generation, as shown in Fig. 7 (the populations of the last generation were additionally considered to examine the relationship between the two objectives through exploring a wider design space near the Pareto solutions. When only Pareto solutions are used for data mining, the extracted design rules may contain minor or even wrong information due to the lack of a dataset). This figure indicates that designers whose interest is to

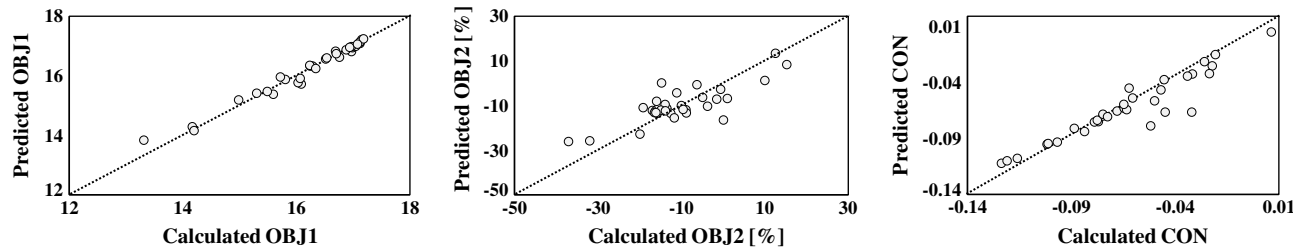


Fig. 6 Validation of converged surrogate model using cross-validation: comparison between calculated value from CFD and predicted value from surrogate model.

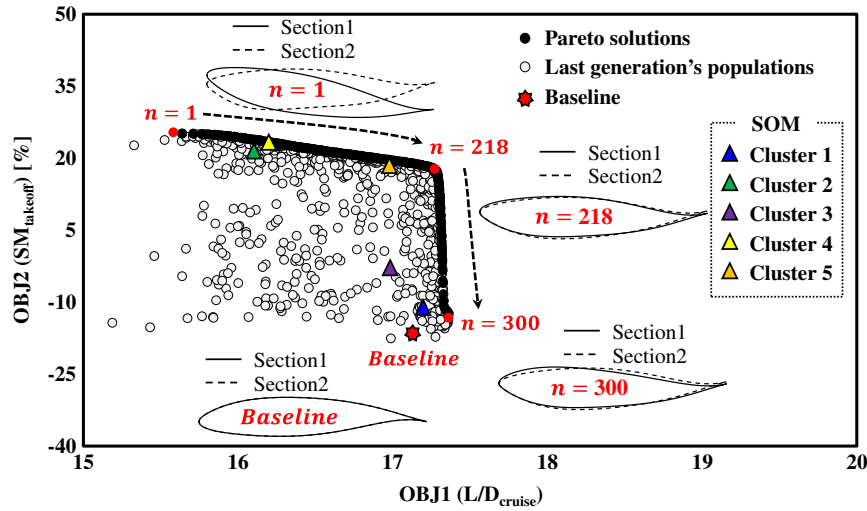


Fig. 7 Sample designs generated based on converged surrogate model: 300 from Pareto solutions ($n = 1$ to $n = 300$) and 981 from the last generation.

increase OBJ2 should be cautious; from $n = 218$ to $n = 1$ along the Pareto solutions, though the increase in OBJ2 is small, the decrease in OBJ1 is relatively large. The same trend can be observed when increasing OBJ1 from $n = 218$ to $n = 300$. These results emphasize the necessity to consider both QOIs simultaneously when designing the UCAV with moderate high-speed performance (OBJ1) and low-speed stability (OBJ2). In the end, the various data mining techniques are applied to the selected sample designs and the overall process for generating this dataset is summarized in Fig. 8.

B. Data Mining Results

1. Univariate Analyses

First, a sensitivity analysis was performed using $n = 218$ from the Pareto solutions (see Fig. 7) as the reference design (since the results of the sensitivity analysis depend on the reference design, the $n = 218$, which was judged to represent the compromise of Pareto solutions, was selected as the reference design). The entire design

space of each design variable was explored in the HK surrogate model, with other design variables fixed to the reference. Considering the relationships between the various design variables and the objective functions, θ_{twist}^1 and θ_{twist}^2 have a noticeable effect on two objectives, as shown in Fig. 9 (each objective is normalized from 0 to 1 based on its minimum and maximum values of the 1281 sample designs used in data mining). Both design variables have a high value of OBJ1 (L/D_{cruise}) at around 0.5, which indicates there the optimal twist angles are almost zero ($\theta_{\text{twist}}^1 = 0$ deg and $\theta_{\text{twist}}^2 = 0$ deg). This is because the higher the twist angle, the higher the drag during cruise flight, which results in a low L/D . They exhibit a similar trend with OBJ2 (SM_{takeoff}): SM_{takeoff} is highest at zero twist angles. Therefore, to achieve both objective functions, section1 and section2 should have almost no twist angle. However, the results of parallel coordinate plot (PCP) in Fig. 10 differ significantly from the sensitivity analysis (PCP only visualizes the top 30% of the design points of each objective function to highlight their distinct trends). In PCP, the higher the value of θ_{twist}^1 , the higher the value of SM_{takeoff} . This result is in agreement with the fact that in Ref. [19], Schütte indicated that increasing the twist angle results in the stronger vortex, which in turn increases the load behind the MRP, reducing the gradient of the pitching moment. The reason why the results of sensitivity analysis and PCP are so different is that univariate sensitivity analysis only considers one varying design variable with respect to the reference design ($n = 218$), while PCP shows a comprehensive relationship between all design variables. Therefore, when the designer only changes the twist angles of the reference design for economic efficiency, designing these angles close to zero can be a reasonable choice, as in sensitivity analysis results. However, note that this cannot be a good choice when other design variables are considered simultaneously, as in PCP results.

Second, the ANOVA was performed using in-house code as in Refs. [47,64,65] on the constructed HK surrogate model for overall design space exploration (the reason why ANOVA was conducted on the overall design space, not for 1281 designs, was that, as previously described, ANOVA requires the integration within the continuous design space: therefore, the uncertainty in the ANOVA results can be higher than in other data mining results, considering that adaptive sampling based on PV and EI is focused on reducing uncertainty near the Pareto solutions rather than the overall design space), and the contribution of each design variable to the variance of each objective function was visualized as a pie chart, as shown in Fig. 11. For OBJ1 (L/D_{cruise}), θ_{twist}^2 has the largest contribution, followed by θ_{twist}^1 . These results are consistent with those of the sensitivity analysis in that the ANOVA can be regarded as a quantitative version of the sensitivity analysis results. For OBJ2 (SM_{takeoff}), the variable X_{low}^1 is the most important and will appear again in the subsequent multi-variate analyses.

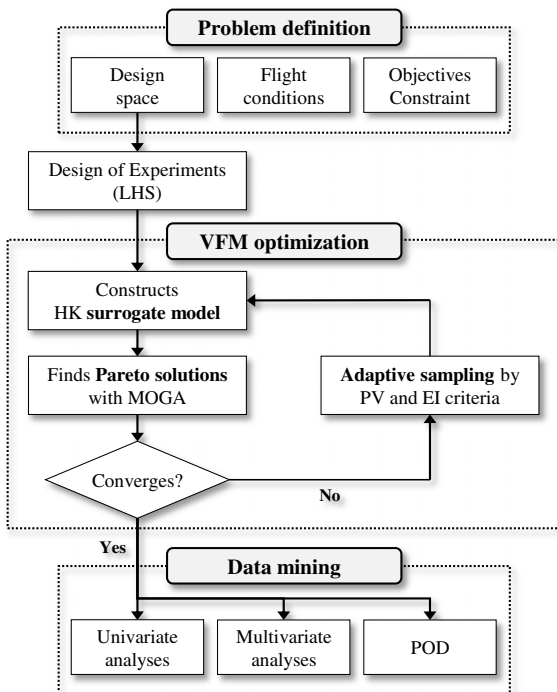


Fig. 8 Flowchart of VFM optimization for generating dataset to be used in data mining.

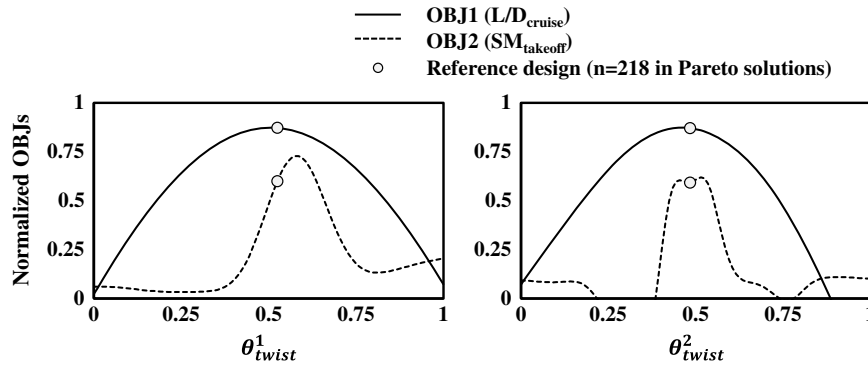


Fig. 9 Sensitivity analysis results of θ^1_{twist} and θ^2_{twist} .

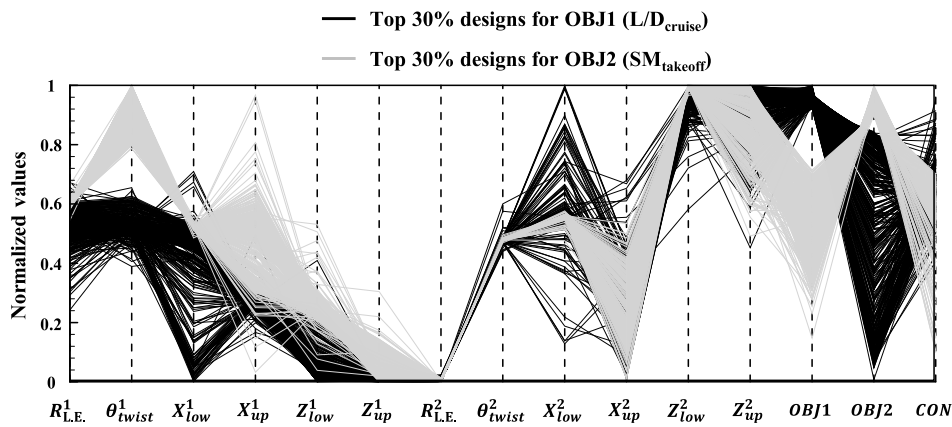


Fig. 10 Parallel coordinate plots of top 30% of designs for two objective functions.

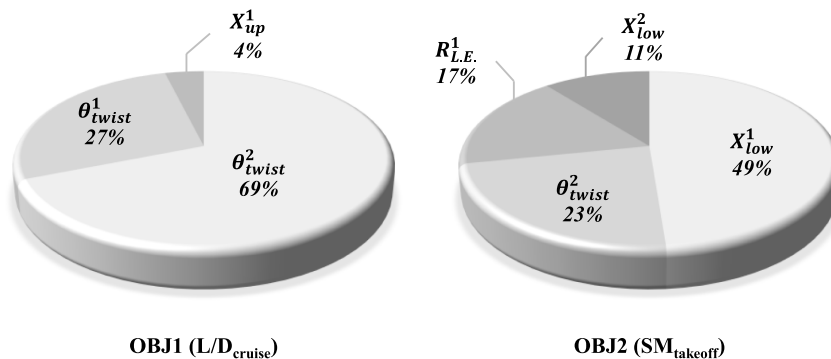


Fig. 11 ANOVA results of each objective function: proportion of each design variable in total variance of objective function.

Finally, the SOM analysis was performed on selected 1281 sample designs to qualitatively evaluate the relationship between the design variables and the objective functions (it was conducted by an open-source tool available on its website[‡]). First, the topology of the output nodes is determined; a hexagonal topology is used herein, enabling the exchange of information with six surrounding nodes. Also, the Euclidean distance is used for calculating the distance between winner and neighboring node d^2_{ci} . A map size of 15×17 was selected for the output nodes, which is appropriate considering the quantization error. Finally, for intuitive visual analysis, the 2-D map was clustered into nine groups. Five characteristic clusters (their average values of the objective functions are shown in Fig. 7) and the SOM results of the objective functions and design variables are shown in Fig. 12 (for the SOM contour, each variable was normalized based on its minimum and maximum values of the 1281 sample designs; red color indicates close to the maximum value, and blue color indicates

close to the minimum value). Comparing the results of the two objectives, they show the opposite color trends in clusters 1–4, which means the two objectives are in a tradeoff relationship in these clusters. Furthermore, in these areas, the design variables θ^1_{twist} , X^1_{up} , Z^1_{low} , and Z^1_{up} have an opposite color trend with OBJ1 (L/D_{cruise}). It indicates that these design variables have a negative correlation with L/D_{cruise} , which can also be predicted from the PCP trends of these design variables (PCP shows that the top 30% of designs for L/D_{cruise} have smaller values of these variables than those for OBJ2 ($SM_{takeoff}$)). As the SOM results are a qualitative analysis, Spearman's rank correlation coefficient is additionally considered in conjunction with the SOM results. This coefficient ρ quantitatively represents monotonic relationships between two variables. The design variables θ^1_{twist} , X^1_{up} , Z^1_{low} , and Z^1_{up} have monotonic decreasing relationships with L/D_{cruise} , with a statistical significance level of over 99.5%; the results are summarized in Table 2. The strongest relationship is between θ^1_{twist} and L/D_{cruise} , and they are visualized by the scatter plot using 1281 samples (Fig. 13). In this figure, the

[‡]<http://livingforsom.com> [retrieved 24 January 2022].

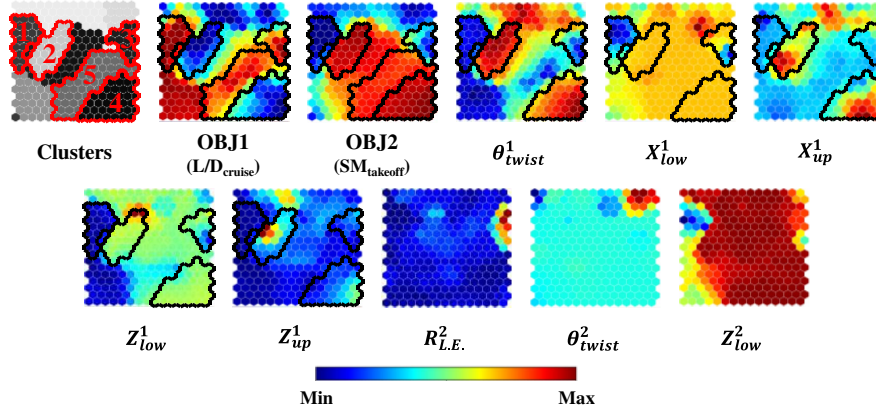


Fig. 12 SOM results for objective functions and design variables (each variable is normalized based on its minimum and maximum values of 1281 designs).

Table 2 Spearman's rank correlation coefficients of each objective function

Objective functions	Design variables	ρ
OBJ1 (L/D_{cruise})	θ^1_{twist}	-0.944
	X^1_{up}	-0.735
	Z^1_{low}	-0.776
	Z^1_{up}	-0.710
OBJ2 (SM_{takeoff})	θ^1_{twist}	0.707

monotonic decreasing relationship is shown apparently. Additionally, from the fact that θ^1_{twist} also has the statistically significant positive correlation with the SM_{takeoff} , it can be concluded that the variable has a major impact on the tradeoff relationship between the two aerodynamic QOIs.

The SOM results of other variables in Fig. 12, $R^2_{L.E.}$, θ^2_{twist} , and Z^2_{low} may seem rather meaningless but paradoxically represent meaningful information: it can be explained by cluster 5 in the SOM results. Herein, both objective functions have high values ($OBJ1_{\text{avg}} = 16.97$, $OBJ2_{\text{avg}} = 18\%$), which means this cluster is a compromise between the two objectives. It contains 254 sample points, and the average values of $R^2_{L.E.}$ and Z^2_{low} are 0 and 0.99, respectively. It indicates that they are at the lower and upper bounds of the design space, respectively, to simultaneously ensure the performance of both objective functions: if $R^2_{L.E.}$ decreases and Z^2_{low} increases above each bound, the performance of the two objective functions can be improved further. As this study is not based on multidisciplinary design, the design space is limited considering the manufacturing and structural requirements of the airfoil. The design

space of the two variables can be expanded in a future study for the further improvement of the two objective functions.

2. Multivariate Analyses

First, a decision tree analysis was performed using all 1281 designs (it was conducted using scikit-learn library [69] of the Python software package). Herein, the classification and regression trees algorithm was selected, which always generates a binary tree. Also, the maximum depth of the tree was set to 4, which is considered sufficient in this case. The results of the decision tree analysis for the objective functions are shown in Fig. 14. For OBJ1 (L/D_{cruise}), the following design rule was extracted:

$$\text{if } \theta^1_{\text{twist}} \leq 0.559 \text{ and } X^1_{\text{up}} > 0.264, \text{ then } OBJ1_{\text{avg}} = 17.272$$

The decision tree analysis results for L/D_{cruise} are consistent with those of the ANOVA in that θ^1_{twist} and X^1_{up} appeared. However, θ^2_{twist} , which was determined to be the most important variable in the ANOVA, does not appear in the decision tree. Unlike the ANOVA that explored the entire design space using the converged surrogate model, the decision tree used the selected 1281 designs. Therefore, the θ^2_{twist} is concentrated in a narrow design space as indicated in the PCP; so that it does not appear in the decision tree results as its algorithm splits the sample designs to maximize the variance of the two groups. Another reason why the results of the two analyses differ is that θ^2_{twist} may have a stronger influence on L/D_{cruise} in the univariate situation compared to the multivariate situation. This will be confirmed later by the results of the rough set theory.

For OBJ2 (SM_{takeoff}), the following design rule was extracted:

$$\text{if } X^1_{\text{low}} > 0.465, R^1_{L.E.} > 0.547 \text{ and } 0.474 < \theta^2_{\text{twist}} \leq 0.482, \\ \text{then } OBJ2_{\text{avg}} = 19.5\%$$

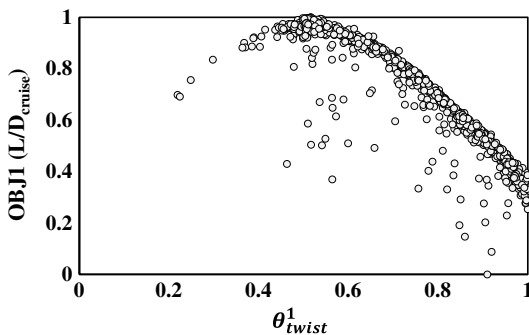


Fig. 13 Scatter plot of 1281 sample designs with respect to the OBJ1 and θ^1_{twist} .

The variables in the decision tree analysis of SM_{takeoff} appear in the order of X^1_{low} , $R^1_{L.E.}$, and θ^2_{twist} , which is similar to the results of the ANOVA. In particular, X^1_{low} appears at the first split, and the average value of SM_{takeoff} is 16.8% when $X^1_{\text{low}} > 0.465$, which is a radical increase in that its average is -5.4% when $X^1_{\text{low}} \leq 0.465$. This is consistent with the SOM results, which indicated that X^1_{low} and SM_{takeoff} are positively correlated. Despite the split in X^1_{low} and the subsequent division of $R^1_{L.E.}$ into a range >0.547 , a total of 967 samples still remained from the initial 1281 points. Subsequently, θ^2_{twist} appears twice in a row and its range is confined to an extremely narrow space (despite the two splits by θ^2_{twist} , the remaining samples only decreased from 967 to 833). This result implies that after the division of X^1_{low} and $R^1_{L.E.}$, the influence of θ^2_{twist} on the performance of SM_{takeoff} becomes significantly dominant. To increase SM_{takeoff} in

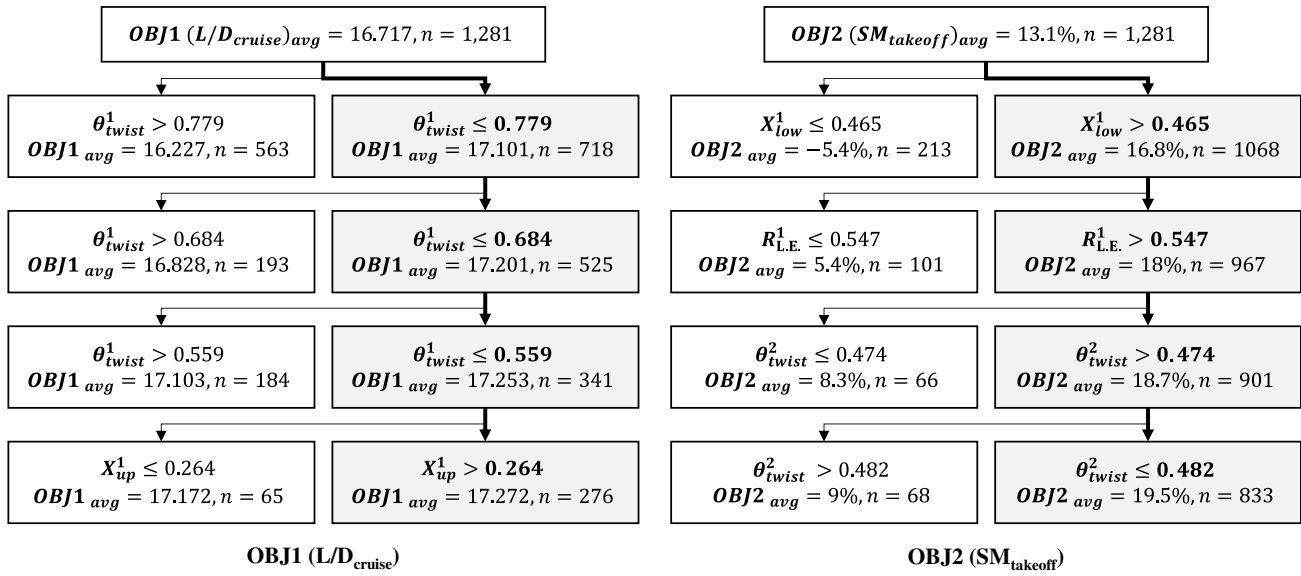


Fig. 14 Decision tree results for each objective function.

	Extracted rules													
DV	1	2	3	4	5	6	7	8	9	10	11	cnt	avg	std
$R_{L.E.}^1$	1	2		1		2		1				5	1.4	0.49
θ_{twist}^1	1	1	1	1	1	1	1	1	1	1	1	11	1	0
X_{low}^1														
X_{up}^1	1							2	2			3	1.7	0.47
Z_{low}^1		1										1	1	0
Z_{up}^1						1		2				2	1.5	0.5
$R_{L.E.}^2$	3			3						3		3	3	0
θ_{twist}^2				1	2		1		2		2	5	1.6	0.49
X_{low}^2			2								2	2	2	0
X_{up}^2			3		3		1					3	2.3	0.94
Z_{low}^2														
Z_{up}^2							2			2		2	2	0
Match	37	32	28	28	27	25	24	24	23	22	21			

a) Rules for OBJ1 (L/D_{cruise}): filtered by max length 4

	Extracted rules								
DV	1	2	3	4	5	6	cnt	avg	std
$R_{L.E.}^1$					1	1	2	1	0
θ_{twist}^1	1	1			1	1	4	1	0
X_{low}^1		1	1	1			3	1	0
X_{up}^1	1	1	1	1	1	1	6	1	0
Z_{low}^1			1				1	1	
Z_{up}^1									
$R_{L.E.}^2$		2	2	2			3	2	0
θ_{twist}^2	1			1			2	1	0
X_{low}^2	1				3	3	3	2.3	0.94
X_{up}^2	1			3	3		3	2.3	0.94
Z_{low}^2									
Z_{up}^2		3	3			3	3	3	0
Match	46	43	43	41	39	33			

b) Rules for OBJ1 (L/D_{cruise}): without filtering

Fig. 15 Rule sets for OBJ1.

the remaining samples, θ_{twist}^2 should be limited to the corresponding design space, which is already revealed by the PCP and the SOM results.

Next, rough set theory was applied for 1281 sample designs (it was conducted by an open-source tool available on its website[§]). As mentioned earlier, the design variables and objective functions are divided into three levels depending on their values: level1 (low), level2 (medium), and level3 (high). Then, Johnson's algorithm, which uses a greedy search to find one reduction per objective function [70], is applied for the reduction process. For comparing the results according to the complexity of multivariate situations, rule sets with filtering (filtered by the maximum length of four, which means that the number of design variables appearing in the IF-part is limited to a maximum of four) and without filtering were shown: the former are design rules in less complex multivariate situations than the latter. The corresponding rules for high values of OBJ1 (L/D_{cruise}) are shown in Fig. 15 (in this figure, "Match" means the number of sample designs corresponding to that design rule, "cnt" means the number of times that the corresponding design variable appears in the extracted rules, "avg" means the average level of the corresponding design variable, and "std" means the standard

deviation of the levels of the corresponding design variable). Note that the rules of length equal to or less than four in Fig. 15b are removed due to their overlap with those in Fig. 15a. Herein, θ_{twist}^1 always appears in filtered results (Fig. 15a); without filtering, the appearance of θ_{twist}^1 decreases, and rather X_{up}^1 appears in all design rules (Fig. 15b). Considering the results of the univariate analysis (ANOVA) that X_{up}^1 (4%) occupied a significantly lesser proportion than θ_{twist}^1 (27%), these multivariate results (rough set) suggest that the impact of X_{up}^1 on L/D_{cruise} increases relative to that of θ_{twist}^1 as the complexity of the interactions between the variables increases. For X_{low}^1 , which has not appeared in the results of the other data-mining methods for L/D_{cruise} , it appears three times in the unfiltered rules, indicating its importance in the multivariate situation. Lastly, in the univariate ANOVA and the multivariate decision tree analysis, the influence of θ_{twist}^2 on L/D_{cruise} was found to decrease with the increase in the complexity of the situation (it was analyzed as the most crucial factor in ANOVA, but it did not appear in decision tree design rules). It is consistent with the rough set theory results as θ_{twist}^2 appears less frequently than other design variables such as θ_{twist}^1 and X_{up}^1 in the rules without filtering. Thus, in section2, the more complex the airfoil shape deformation is, the weaker the effect of the twist angle on the flow.

[§]<http://bioinf.icm.uu.se/rosetta> [retrieved 24 January 2022].

	Extracted rules												
DV	1	2	3	4	5	6	7	8	9	10	cnt	avg	std
R^1_{LE}	2		2		2			2			4	2	0
θ^1_{twist}	3	3	3	3	3		3	3	3	3	9	3	0
X^1_{low}	2	2				3		2	2		5	2.2	0.4
X^1_{up}								2			1	2	0
Z^1_{low}						3	3				2	3	0
Z^1_{up}	2		2		2						3	2	0
R^2_{LE}			2	2		1			2		4	1.8	0.43
θ^2_{twist}		1				2	1		1	1	5	1.2	0.4
X^2_{low}				3						1	2	2	1
X^2_{up}				1							1	1	0
Z^2_{low}					3						1	3	0
Z^2_{up}		1					1			1	3	1	0
Match	38	34	21	21	19	18	17	16	16	16			

a) Rules for OBJ2 (SM_{takeoff}): filtered by max length 4

	Extracted rules										
DV	1	2	3	4	5	6	7	8	cnt	avg	std
R^1_{LE}	2	2	2	2	2	2			6	2	0
θ^1_{twist}	3		3	3	3	3	3	3	7	3	0
X^1_{low}	2			3	2	2	2	2	6	2.2	0.37
X^1_{up}	3		3						2	3	0
Z^1_{low}							3		1	3	0
Z^1_{up}											
R^2_{LE}		2		2	1	2	2	2	6	1.8	0.37
θ^2_{twist}	2	2	2	2	2	2	2		7	2	0
X^2_{low}		2	2						2	2	0
X^2_{up}								1	1	1	0
Z^2_{low}		3	3					3	3	3	0
Z^2_{up}											
Match	40	30	30	25	24	23	23	21			

b) Rules for OBJ2 (SM_{takeoff}): without filtering

Fig. 16 Rule sets for OBJ2.

The extracted rules for high values of OBJ2 ($SM_{takeoff}$) are shown in Fig. 16. R_{LE}^1 , X_{low}^1 , and θ_{twist}^2 appear frequently in both cases, and considering they also appeared in the ANOVA and decision tree results, their significant influence on $SM_{takeoff}$ can be identified. However, unlike the previous analyses, θ_{twist}^1 appears to have significant importance in the rough set theory results. It appears nine times in Fig. 16a and seven times in Fig. 16b. In particular, it has consistently high values of level3, which implies that θ_{twist}^1 should be high for low-speed longitudinal stability (this is also reflected in the PCP). Furthermore, given that it also frequently appeared as a level1 value in the rough set results of OBJ1, it can be concluded that θ_{twist}^1 plays a crucial role in the tradeoff relationship between the two objective functions.

The previous discussions investigate the effects of the airfoil shapes of two sections of the UCAV configuration on the aerodynamic performance during cruise and the longitudinal stability during low-speed flight. Through various data mining techniques, the comparison between univariate analysis results and multivariate analysis results (in particular, the rough set theory allows comparisons based on the degree of multivariate situations) was conducted, and different design rules were established depending on how complex the interactions between the design variables are considered. The univariate analysis can be practical in that it offers reliable results in the situation where the engineer scrutinizes the effect of the single design variable. However, in real engineering problems, this situation is uncommon: multiple design variables influence each other in a complex way (this study also corresponds to a multivariate design in which 12 design variables were considered simultaneously). Therefore, univariate

analysis, which cannot account for it can lead to wrong results, whereas the multivariate analysis offers reliable results. This underlines the need to perform both univariate and multivariate techniques to analyze the UCAV design accurately and efficiently.

Regardless of the analysis method, θ_{twist}^1 appeared frequently in the data mining results (sensitivity analysis: it should be almost equal to zero to for both objective functions, ANOVA: it accounts for 27% of the variance in L/D_{cruise} , SOM: it has the negative correlation with L/D_{cruise} and positive correlation with $SM_{takeoff}$, decision tree: its lower value is suggested for L/D_{cruise} by appearing three times in a row, and rough set theory: it should have lower values for L/D_{cruise} and higher values for $SM_{takeoff}$). Except for the sensitivity analysis results, they consistently show that θ_{twist}^1 should have a lower value for L/D_{cruise} and a higher value for $SM_{takeoff}$. To further analyze the effects of θ_{twist}^1 , the flow of the $n = 1$ (which offers excellent $SM_{takeoff}$ performance with θ_{twist}^1 as 6.99 deg) and $n = 300$ (which offers excellent L/D_{cruise} performance with θ_{twist}^1 as 0.22 deg) design points in Fig. 7 were visualized in Fig. 17. The surface pressure contour, streamlines, and iso-surface of $V_x = 0$ (green-colored, reverse flow region due to vortex breakdown) during take-off are also shown. For $n = 1$ design, the vortex topology behind the crank (upper side of section1) is composed of the apex vortex and the crank vortex. However, for $n = 300$, only the apex vortex affects the formation of the vortex topology behind the crank. The crank vortex is generated in the $n = 1$ design due to the large θ_{twist}^1 and the corresponding high effective AOA, which strengthens the resultant vortical flow on the upper side of the wing. This property is also proven in Fig. 17c, where $n = 1$ has a higher suction at the front of the

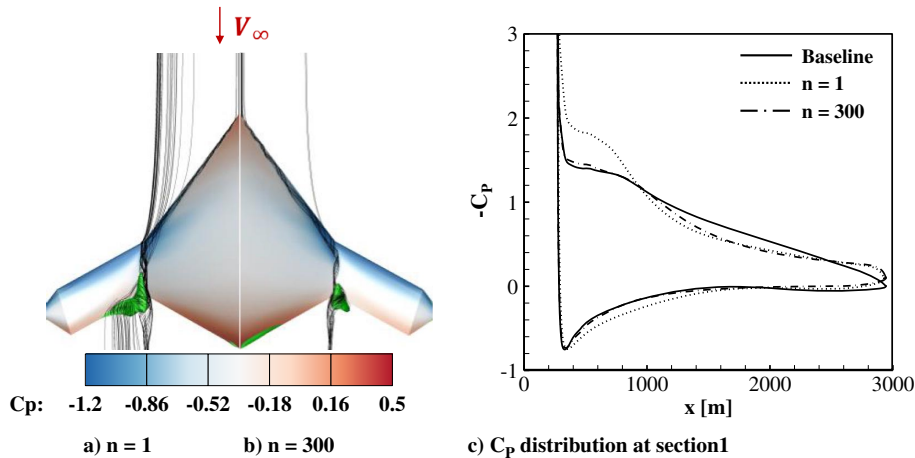


Fig. 17 a-b) Upper surface pressure, streamlines, and iso-surface of $V_x = 0$ at takeoff for $n = 1$ and $n = 300$ designs; c) C_p distribution at section1.

upper airfoil at section1 than the other designs. Thus, owing to the stronger suction behind the MRP, the $n = 1$ configuration experiences a strong nose-down moment, which ensures the good SM_{takeoff} performance. However, the L/D_{cruise} of $n = 1$ is inferior to that of $n = 300$, owing to its excessive twist angle. Indeed, for $n = 300$, the twist angle is almost equal to 0 deg, as favored by the previous sensitivity analysis. In this regard, the sensitivity analysis provided appropriate design rules for L/D_{cruise} , but not for SM_{takeoff} , in terms of θ_{twist}^1 .

3. Airfoil Shape Visualization by Proper Orthogonal Decomposition

The data mining methods employed so far were based on the selected PARSEC parameters. However, these analyses have the following limitations: 1) extracted design rules cannot be used with different airfoil parameterization methods; and 2) it is difficult to visually identify the required airfoil deformations for the improvement of each objective function's performance. For example, when the airfoils are parameterized using B-spline control points, the design rules extracted herein cannot be used as reference as they are based on PARSEC parameters. Furthermore, it is difficult to visualize the camber and thickness distributions for the performance of each objective function. Therefore, data mining based on POD (scikit-learn library [69] of the Python software package was used) was conducted to overcome these limitations. Universal design rules

were extracted that can be referred to regardless of the airfoil parameterization method, visualizing the airfoil camber and thickness distribution required to improve the performance of each objective function. POD used only the Pareto solutions to focus on the tradeoff relationship between the two objectives, unlike the other data mining techniques (SOM, decision tree, and rough set theory) that used both the Pareto solutions and the last generation populations. It was performed separately for section1 and section2 airfoil, and the number of modes to be used for dimensional reduction was determined based on the energy ratio calculated using Eq. (10).

The airfoil at section1 was analyzed first. The top two modes (first and second mode) preserve about 95% of the total energy, and the components of these modes with respect to the Pareto solution numbering (n) are shown in Fig. 18 (the Pareto solution numbering (n) are shown in Fig. 7; smaller values of n indicate a higher priority for OBJ2 performance and larger values indicate a higher priority for OBJ1 performance). As shown in Fig. 18, the smaller the value of n , the more dominant the second mode. It implies that the airfoil shape represented by the second mode should be prominent for the OBJ2 performance. Conversely, the first mode should be prominent for OBJ1. The corresponding camber and thickness distributions of each mode are visualized in the insets of Fig. 18 (Figs. 18a and 18b). In summary, a higher value of n improves the performance of OBJ1 (L/D_{cruise}), which means that the camber and thickness distributions

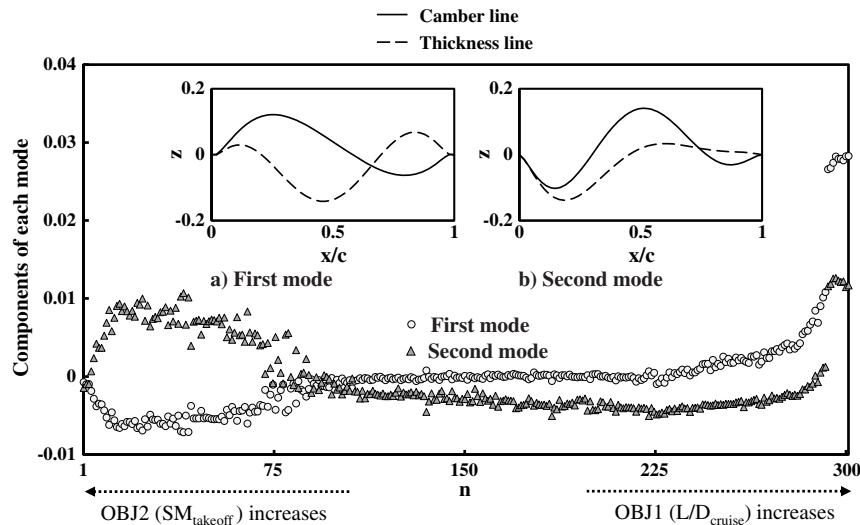


Fig. 18 POD components of two dominant modes at section1 with respect to n ; camber and thickness distributions of these modes (insets).

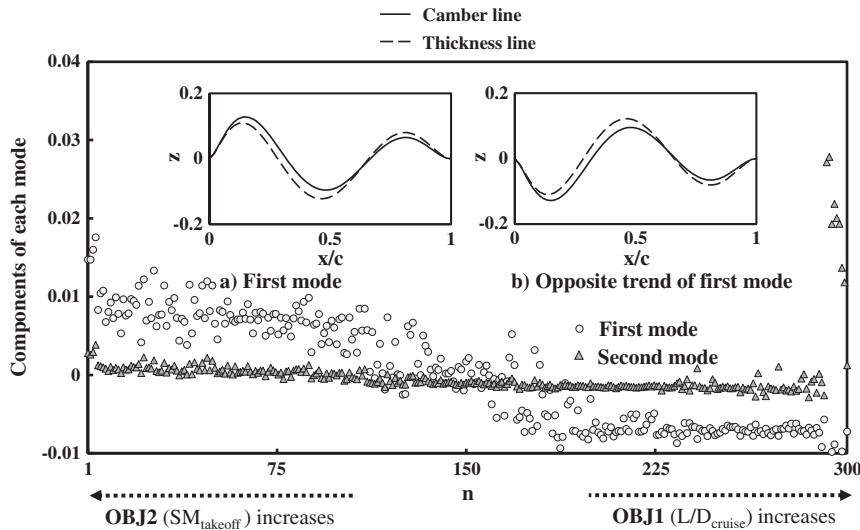


Fig. 19 POD components of two dominant modes at section2 with respect to n ; camber and thickness distributions of first mode and its opposite trend (insets).

should follow the trend in the first mode: the camber and thickness near the leading edge and the thickness near the trailing edge should increase. Likewise, to increase OBJ2 (SM_{takeoff}), they should take the trend of the second mode: decreased camber and thickness near the leading edge and increased camber at the midpoint.

Next, section2 was investigated. As in section1, the first two modes preserved the 92% of the total energy, and their components are shown in Fig. 19. Contrary to the section1, the components of the second mode are close to zero regardless of the value of n : it means that the second mode of section2 has minor effects on the tradeoff relationship between two objective functions, though the total energy preservation rate was only 72% with the first mode alone. Therefore, only the first mode was analyzed. Its positive components become dominant as n decreases, which indicates that the trend of the first mode should be reflected in the section1 airfoil shape for the SM_{takeoff} performance. Accordingly, as shown in Fig. 19a, both the camber and thickness should increase near the leading and trailing edges and decrease near the midpoint. In contrast, the negative components become dominant (which means the opposite trend of the first mode) as n increases. Therefore, in situations where L/D_{cruise} is the primary focus of UCAV design, both the camber and thickness should decrease near the leading and trailing edges and increase near the midpoint as shown in Fig. 19b (note: the opposite trend of the first mode is equivalent to its symmetry about the x axis).

IV. Conclusions

This study suggested the efficient and accurate framework for the design rule extraction of the UCAV configuration. As the previous studies primarily focused on the low-speed longitudinal stability due to the unstable pitching-moment characteristic, the aerodynamic performance of the high-speed cruise flight was not considered. Also, to scrutinize the intricate relationships between design variables and QOI, univariate analyses based on low-fidelity flow solvers were conducted due to the limitations of the computational resources. Herein, both low-speed and high-speed flight conditions were taken into account through the multi-objective optimization process: the first objective function as the high-speed aerodynamic performance and the second as the low-speed longitudinal stability. In addition, the high-fidelity flow solver RANS was adopted to precisely capture the vortical flow structure, which is the main cause of instability of the pitching moment. Consideration of two flight conditions with high-fidelity flow analysis requires enormous computing cost compared to the previous studies, and the corresponding cost was mitigated by multi-fidelity surrogate modeling. Based on the constructed surrogate model, sample designs were generated and various data mining techniques were applied to extract the complex but realistic design rules of the UCAV.

The data mining techniques employed to establish the design rules of the two aerodynamic QOI were divided into three main categories: univariate analyses, multivariate analyses, and airfoil shape visualization using POD. The simple and straightforward univariate analyses were conducted first, which can suggest erroneous results in multivariate situations. Then, the multivariate analyses were applied to extract complex and intricate, but reliable design rules. At last, to represent the design rules more intuitively, required airfoil shape

deformations for the improvement of each QOI were presented visually using POD. Finally, the following insights were obtained:

1) The design rules extracted from univariate and multivariate analyses can have conflicting results. For example, univariate sensitivity analysis indicated that θ_{twist}^1 should be almost equal to zero for the performance of both QOIs. However, rough set theory (multivariate analysis) derived different design rules, which were verified to be more realistic through flowfield analysis: θ_{twist}^1 should decrease for L/D_{cruise} performance and increase for SM_{takeoff} . These results support the motivation of this study to perform multivariate analyses (nevertheless, note that univariate and multivariate analyses can also yield consistent rules; the positive correlation of X_{low}^1 with SM_{takeoff} was discovered in both SOM and decision tree results).

2) The design variable θ_{twist}^1 was found to be the most important factor in the tradeoff relationship between two QOIs: an increase in θ_{twist}^1 strengthens the vortex behind the MRP, resulting in a nose-up pitching moment that increases SM_{takeoff} , but decreases L/D_{cruise} due to the excess effective angle of attack.

3) The universal design rules of the airfoils at two sections were visualized. They can be referred to regardless of the parameterization method used.

This time-efficient and realistic design rule extraction framework for the UCAV configuration and the approaches to analyzing extracted design rules from the various data mining techniques presented in this study can be extended to any engineering field.

Appendix: Flow Solver Validation

As mentioned in the main text, the flow solver validation comparing C_L , C_D and C_M with respect to AOA was also conducted (Fig. A1). Their results were almost the same as the CFD simulations using the cell-vertex approach and the Wilcox- $k-\omega$ turbulence model in Ref. [14]. Lift and drag coefficients were calculated accurately, and the pitching-moment coefficients were somewhat underestimated but accurately captured the pitch break location when compared to the experimental data, as stated in Ref. [14].

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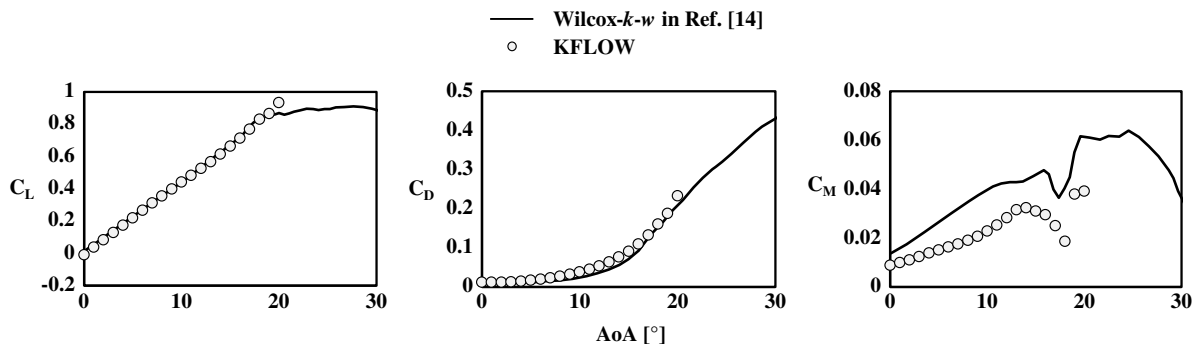


Fig. A1 Comparison of aerodynamic coefficients at $Ma = 0.15$, $Re = 1.6 \times 10^6$ between CFD results in Ref. [14] and KFLOW.

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